ARAS-FCWNet: Low-Cost and Lightweight Model for Forward Collision Warning (FCW) Systems in Unstructured ARAS Environments

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Abstract

Advanced Riding Autonomous Systems (ARAS) are critical for enhancing safety in modern vehicles, including motorcycles and bicycles. A Forward Collision Warning (FCW) system is a crucial component of ARAS that aims to prevent accidents by alerting the rider to imminent collisions. However, existing FCW systems often rely on multiple high-resolution sensors and sophisticated hardware, which are cost-prohibitive and computationally intensive for widespread adoption, especially in low-cost vehicles. Developing an effective and affordable FCW system for ARAS poses several technical challenges, including sensor limitations, computational constraints, real-time performance, environmental variability, and lightweight algorithms. We propose a low-cost, lightweight perception model optimized for a 1.3 MP RGB camera supplemented by IMU and vehicle odometry. The framework accurately detects potential collisions in real time with minimal false-positive and false-negative rates. We optimize the system for resource-constrained embedded platforms by balancing processing speed, memory footprint, and power consumption. Extensive validation across diverse environments demonstrates the robustness and reliability of the proposed FCW solution for low-cost two-wheeler safety.

Keywords: Advanced Rider Assistance System (ARAS), Forward Collision Warning (FCW), Unstructured-ARAS, RGB Pinhole-Camera, Low-cost Lightweight Model.

1 Introduction

Advanced Rider Assistance Systems (ARAS) are gaining significant attention for improving safety in motorcycles, scooters, bicycles, and other lightweight mobility [Kaye et al., 2024]. Compared with conventional four-wheeler Advanced Driver Assistance Systems (ADAS), ARAS operates under more challenging conditions due to rider instability, narrow vehicle geometry, irregular traffic behavior, and highly unstructured road environment platforms [Naweed and Blackman, 2024]. Among different ARAS functionalities, Forward Collision Warning (FCW) accurately forecasting the trajectories of road users remains highly challenging, because of the inherent uncertainty in human behavior, environmental variability, occlusions, and heterogeneous interactions among multiple dynamic agents, as explained in Fig 1. incorporating radar, LiDAR, stereo cameras, surround-view systems, and high-resolution perception sensors [**?**]. Although these approaches achieve reliable performance, they significantly increase

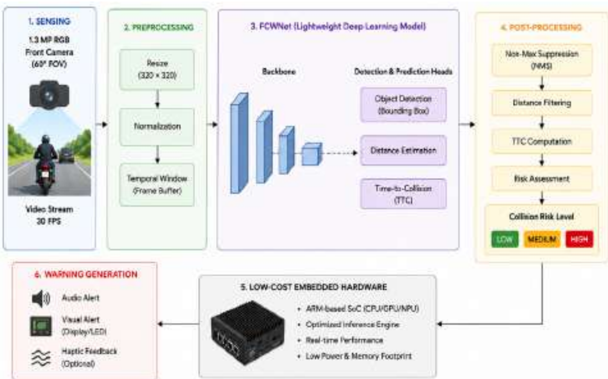

Figure 1: Conceptual Diagram of ARAS-FCWNet

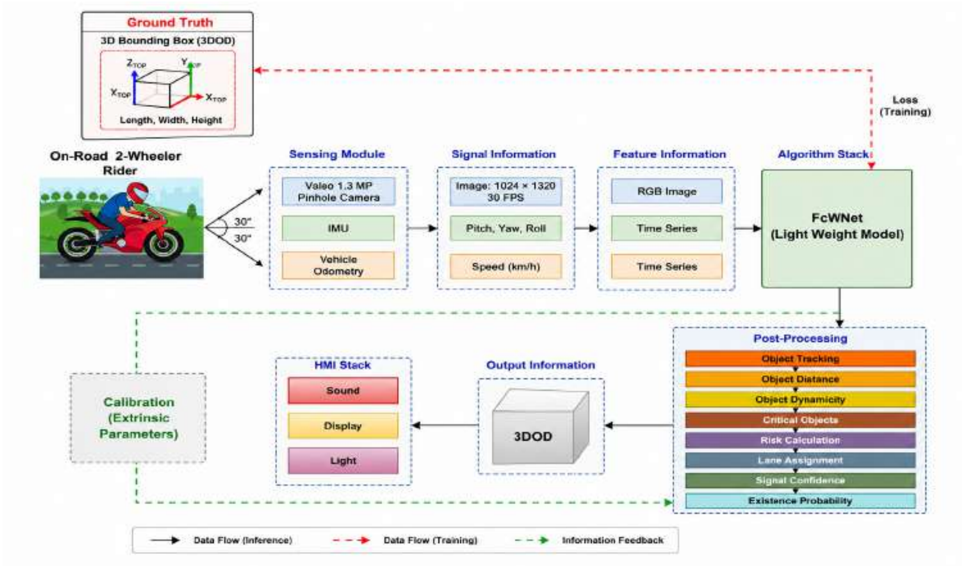

Figure 2: Proposed ARAS-FCWNet Framework.

hardware cost, computational complexity, memory consumption, and power requirements. Such solutions are often unsuitable for low-cost ARAS deployment, particularly in emerging markets where affordable two-wheeler safety systems are essential.

Recent advances in deep learning and computer vision have enabled the development of vision-based FCW systems using object detection and scene understanding frameworks such as Faster R-CNN, SSD, YOLO, and monocular 3D object detection approaches [Manakitsa et al., 2024], [Zhang et al., 2025]. Several lightweight perception models including MobileNet [Howard et al., 2017], EfficientNet [Tan and Le, 2019], and Tiny-YOLO [Redmon and Farhadi, 2018] have also been proposed for embedded deployment. However, most existing methods are designed for structured road environments and depend heavily on lane markings, high-resolution sensors, or powerful GPU-based [Mihalj et al., 2022]. Their performance often degrades in unstructured ARAS scenarios involving mixed traffic, vulnerable road users, occlusions, poor illumination, adverse weather, and irregular road boundaries.

Recent studies further explored graph-based and temporal motion-aware frameworks for intelligent transportation systems, incorporating object tracking, trajectory estimation, and collision risk [Wu et al., 2023]. Nevertheless, many existing FCW approaches focus primarily on object detection while neglecting integrated free-space segmentation, depth estimation, motion understanding, and risk-aware reasoning within a unified lightweight architecture [Zhang et al., 2025]. In addition, deploying such systems on low-power embedded hardware while maintaining real-time performance remains a significant challenge.

To address these limitations, this paper proposes *ARAS-FCWNet*, a low-cost, lightweight forward collision warning framework designed specifically for unstructured ARAS environments (some scenarios are shown in Fig. 3). As illustrated in Fig. 2, the proposed system utilizes a single 1.3 MP front-view RGB pinhole camera integrated with IMU and vehicle odometry signals to perform efficient scene perception and collision risk estimation. The framework adopts a lightweight multi-task deep learning architecture capable of jointly performing object detection, free-space estimation, monocular depth prediction, motion estimation, and collision risk assessment in real time.

Unlike conventional FCW systems that depend on expensive multi-sensor setups [Hassan et al., 2026], ARAS-FCWNet operates entirely on monocular vision and low-cost embedded hardware while maintaining efficient inference capability. The framework further integrates object tracking, lane assignment, object dynamicity estimation, critical object identification, and Time-To-Collision (TTC)-based risk calculation through a lightweight post-processing pipeline. Experimental validation under diverse traffic and environmental conditions demonstrates the effectiveness of the proposed approach for real-time ARAS deployment.

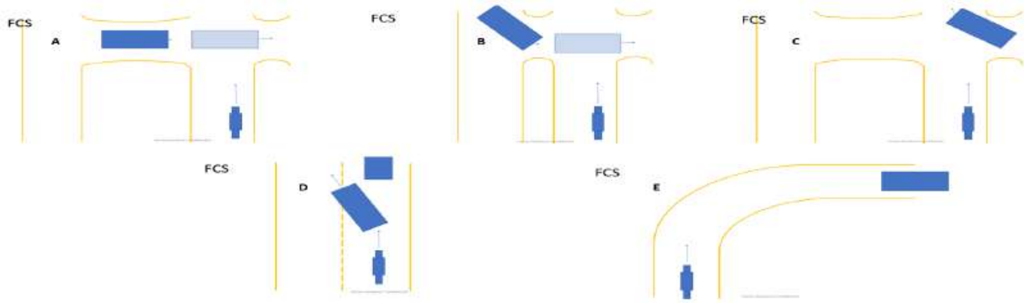

Figure 3: ARAS Scenarios: (A) A stationary car pulls into the motorcycle's path at a divided highway/stop sign crossing. (B) A stationary car enters the motorcycle's path while crossing a divided highway. (C) A parked car suddenly moves into the motorcycle's lane. (D) A following car changes lanes abruptly to avoid a suddenly stopped vehicle, while the motorcycle never observes the stopped vehicle moving. (E) A motorcycle rounds a curve as a car ahead slows to a stop in the same lane.

The primary contributions of this work are summarized as follows:

- A lightweight monocular FCW framework for unstructured ARAS environments using a single low-cost RGB camera.
- A unified multi-task deep learning architecture integrating object detection, free-space segmentation, depth estimation, and motion prediction.
- A computationally efficient embedded perception pipeline optimized for real-time ARAS deployment on low-power hardware platforms.
- A robust collision risk estimation framework incorporating object tracking, TTC analysis, and rider-centric warning generation.

2 Proposed Methodology

The proposed *ARAS-FCWNet* framework is designed as a lightweight monocular Forward Collision Warning (FCW) system for unstructured Advanced Rider Assistance System (ARAS) environments. As illustrated in Fig. 2, the framework utilizes a single low-cost 1.3 MP RGB front-view camera along with IMU and vehicle odometry signals to perform efficient collision risk estimation in real time. The overall architecture consists of five major stages: (i) image acquisition and preprocessing, (ii) lightweight multi-task perception, (iii) object tracking and motion estimation, (iv) collision risk analysis, and (v) rider warning generation.

2.1 Input Acquisition and Preprocessing

Let the monocular RGB image captured at time instant t be represented as:

$$I_t \in \mathbb{R}^{H\times W\times 3} \tag{1}$$

where H and W denote the image height and width, respectively. The captured frame is resized and normalized before being forwarded to the proposed FCWNet backbone. Simultaneously, inertial and odometry measurements are collected:

$$S_t = \{v_t, a_t, \omega_t\} \tag{2}$$

where v_t denotes vehicle velocity, a_t denotes acceleration, and ω_t represents yaw rate obtained from IMU sensors.

2.2 Lightweight Multi-Task Perception Network

ARAS-FCWNet uses a lightweight MobileNetV3-Small backbone (2.5M parameters, 0.42 GFLOPs) operating at a unified input resolution of 320×320 pixels. The shared backbone extracts hierarchical features feeding into task-specific decoders for 2D bounding box detection, free-space segmentation, monocular depth prediction (via pinhole camera projection constraints $d = f \cdot Y / y$), and temporal motion analysis. This unified multi-task design achieves a memory footprint under 18 MB on embedded devices.

2.3 Distance Estimation and Dynamic Object Analysis

Object distance is estimated using monocular geometric constraints based on object scale variation in image space. Relative velocity is computed from temporal distance changes between consecutive frames. Lightweight temporal association is further employed for object tracking and motion-state estimation.

The object dynamicity score is defined as:

$$\Gamma_i = \sqrt{(v_{x,i})^2 + (v_{y,i})^2} \tag{3}$$

where $v_{x,i}$ and $v_{y,i}$ denote horizontal and vertical motion components, respectively.

2.4 Critical Object Selection and Collision Risk Estimation

Collision priority is determined using object distance, motion dynamicity, and detection confidence:

$$C_i = \alpha \frac{1}{d_i} + \beta \Gamma_i + \gamma c_i, \tag{4}$$

where the weighting coefficients are empirically tuned as $\alpha = 0.5$, $\beta = 0.3$, and $\gamma = 0.2$.

Collision risk is estimated using Time-To-Collision (TTC), and the final warning score is computed as:

$$R_i = \exp(-\lambda \cdot TTC_i), \tag{5}$$

where the sensitivity parameter is set to $\lambda = 1.2$.

An FCW alert is triggered whenever the estimated collision risk exceeds a predefined threshold.

2.5 Joint Optimization

The proposed framework is trained end-to-end using a unified multi-task objective that jointly optimizes detection, segmentation, depth estimation, and collision-risk prediction:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{det} + \lambda_2 \mathcal{L}_{seg} + \lambda_3 \mathcal{L}_{depth} + \lambda_4 \mathcal{L}_{risk}, \tag{6}$$

where the loss weights are selected through validation tuning as $\lambda_1 = 1.0$, $\lambda_2 = 0.5$, $\lambda_3 = 0.5$, and $\lambda_4 = 0.2$.

The lightweight design enables efficient real-time deployment on resource-constrained ARAS hardware while maintaining reliable collision warning performance in unstructured traffic environments.

3 Experimental Results

The proposed ARAS-FCWNet framework is implemented using PyTorch and evaluated on a workstation equipped with a 13th Gen Intel Core i7-13850HX processor, 64 GB RAM, and a 4 GB GPU. The model is trained using the Adam optimizer with an initial learning rate of 1×10^{-4} and a batch size of 64. All experiments converge within approximately 45 epochs. To ensure suitability for embedded ARAS deployment, the proposed network is optimized for low memory consumption and real-time inference.

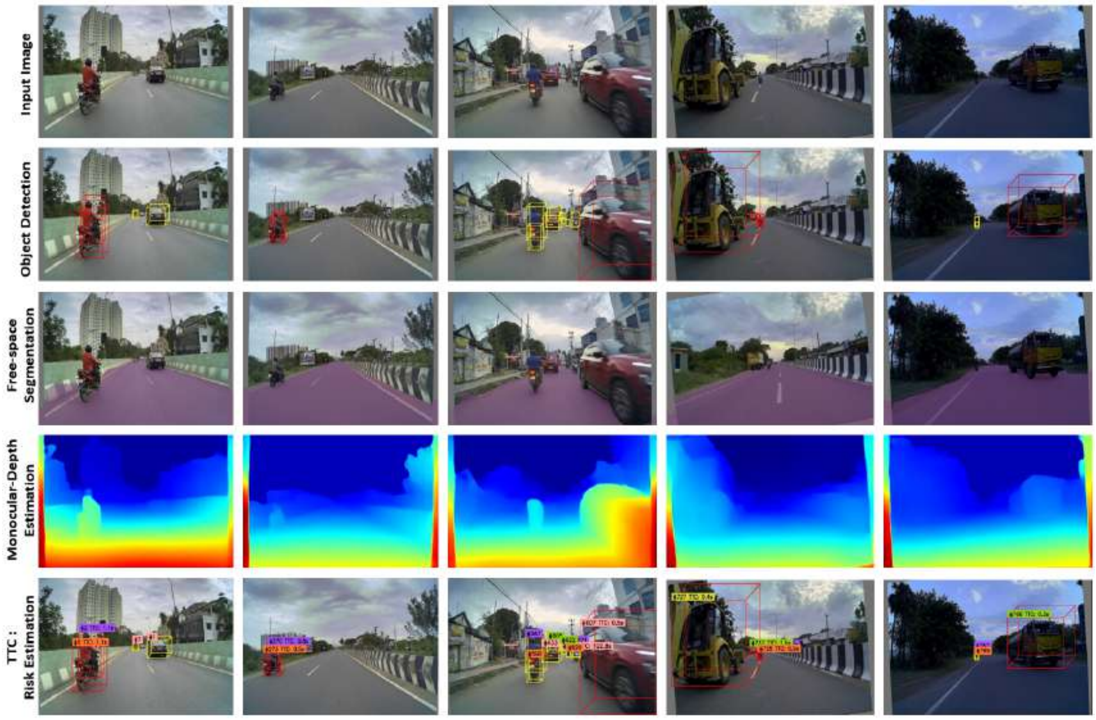

Figure 4: ARAS-FCWNet Multi-Task Perception Network: (i) Object Detection, (ii) Free-space Segmentation, (iii) Monocular-Depth Estimation, and (iv) TTC-based Risk Estimation.

3.1 Dataset and Experimental Setup

ARAS-FCWNet is trained and tested across BDD100K [Yu et al., 2020] (10,000 frames), nuScenes [Caesar et al., 2020] (5,000 keyframes), and a proprietary Valeo ARAS dataset (12,000 frames captured from rider-perspective 1.3 MP cameras across unstructured Indian traffic). Four-wheeler datasets (BDD100K/nuScenes) provide rich front-view traffic dynamics, while the Valeo dataset supplies true two-wheeler lean dynamics and irregular road layouts. Experiments are averaged across 5 independent runs (70:10:20 split).

3.2 Evaluation Metrics

The performance of ARAS-FCWNet is evaluated using standard FCW and perception metrics, including Precision, Recall, F1-score, Mean Average Precision (mAP), False Positive Rate (FPR), False Negative Rate (FNR), and inference latency. In addition, Time-To-Collision (TTC) estimation accuracy and collision warning accuracy are also analyzed.

The Mean Average Precision (mAP) is used to evaluate object detection performance, while the Time-To-Collision (TTC) ground truth is computed using:

$$TTC_{gt} = \frac{d_{rel}}{v_{rel}}, \tag{7}$$

where d_{rel} and v_{rel} denote the relative distance and relative velocity between the ego vehicle and the target object, respectively. The TTC ground truth was generated on the Valeo test split using calibrated 3D bounding boxes obtained from reference LiDAR/radar annotations.

3.3 Quantitative Performance Comparison

For a fair comparison, all baseline models were retrained from scratch using the same unified dataset split and identical training protocol, with an IoU threshold of 0.50 for object detection evaluation. The proposed ARAS-

FCWNet is compared against lightweight real-time perception models, including Tiny-YOLO, MobileNet-SSD, and YOLOv5-Nano. Table 1 reports the combined test performance across the BDD100K, KITTI, and Valeo datasets, demonstrating the effectiveness of the proposed framework under diverse driving scenarios.

Table 1: Performance comparison of FCW models.

Method	mAP↑	F1-score↑	FPR↓	FPS↑
Tiny-YOLO	71.2	74.5	12.6	42
MobileNet-SSD	75.8	78.4	10.2	38
YOLOv5-Nano	81.5	83.1	8.7	34
ARAS-FCWNet	**85.7**	**87.4**	**6.1**	**48**

The results demonstrate that the proposed ARAS-FCWNet achieves superior detection accuracy while maintaining high real-time performance suitable for low-cost embedded ARAS deployment.

3.4 Collision Warning Performance

To evaluate collision warning capability, TTC-based warning accuracy is analyzed under different riding conditions (debug screen status result of a sample scenario shown in Fig 5). Table 2 presents the TTC estimation and FCW accuracy results.

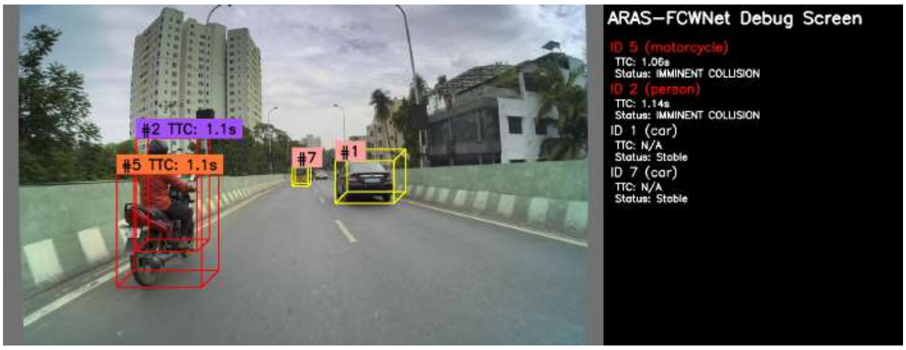

Figure 5: ARAS-FCWNet Debug Screen Results of a Sample Input-Image using TTC Performance-based Estimation of Collision Risk Status (stable in this Scenario)

Table 2: Collision warning evaluation results.

Method	TTC Error↓	Warning Accuracy↑	Latency (ms)↓
Tiny-YOLO	1.42	78.5	41
MobileNet-SSD	1.18	81.2	39
YOLOv5-Nano	0.96	85.7	35
ARAS-FCWNet	**0.71**	**91.3**	**28**

The proposed framework achieves the lowest TTC estimation error and highest warning accuracy, indicating reliable collision risk prediction in unstructured riding environments.

Fig. 4 illustrates qualitative FCW examples generated by ARAS-FCWNet under challenging traffic conditions. The framework successfully detects vehicles, pedestrians, and vulnerable road users while accurately identifying free-space regions and collision threats. The proposed lightweight architecture maintains stable performance under illumination changes, partial occlusions, and dense traffic scenarios.

3.5 Ablation Analysis

An ablation study is conducted to evaluate the contribution of individual modules within ARAS-FCWNet. Table 3 summarizes the effect of removing specific components.

Table 3: Ablation study of ARAS-FCWNet components.

Configuration	mAP↑	Warning Accuracy↑
Without Depth Estimation	80.6	84.5
Without Motion Estimation	81.2	85.1
Without Free-space Segmentation	79.8	83.7
Full ARAS-FCWNet	**85.7**	**91.3**

Figure 6: ARAS-FCWNet Debug Screen Status: Imminent Collision using TTC Performance

The results confirm that integrating monocular depth estimation, motion reasoning, and free-space segmentation significantly improves the robustness of collision prediction and the reliability of FCW.

3.6 Discussion

To evaluate embedded viability, ARAS-FCWNet was deployed on an NVIDIA Jetson Orin Nano (8GB). Operating under an 8W power budget, the TensorRT-optimized engine achieved 42 FPS with a latency of 23.8 ms, consuming 1.2 GB RAM. This confirms real-time embedded performance on low-cost ARM/GPU architectures. Unlike conventional FCW systems relying on expensive multi-sensor fusion, the proposed framework achieves reliable collision warning performance using only a low-cost monocular RGB camera and a lightweight deep learning architecture. The proposed system therefore provides a scalable and affordable solution for next-generation rider assistance systems operating in challenging unstructured traffic environments.

4 Conclusion and Future Scope

This paper presented *ARAS-FCWNet*, a lightweight multi-task perception framework for real-time Forward Collision Warning (FCW) in unstructured ARAS environments. The proposed model integrates multi-task perception with collision-risk assessment using single RGB camera input tightly coupled with vehicle IMU and odometry telemetry. By replacing heavy multi-sensor stacks with fused monocular-inertial cues, ARAS-FCWNet provides an affordable safety framework for two-wheelers in unstructured traffic.

Experiments on BDD100K, nuScenes, and in-house Valeo ARAS datasets demonstrate robust performance across diverse traffic, illumination, and occlusion conditions, while maintaining real-time capability suitable for resource-constrained platforms.

Future work will focus on improving robustness under extreme conditions such as severe occlusions, night-time, and adverse weather using transformer-based temporal modeling and uncertainty-aware risk estimation. Extensions will also explore trajectory-aware prediction, rider intention modeling, and multi-sensor fusion with radar/IMU for enhanced safety in dynamic traffic scenarios.

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